AI Coding for Geo

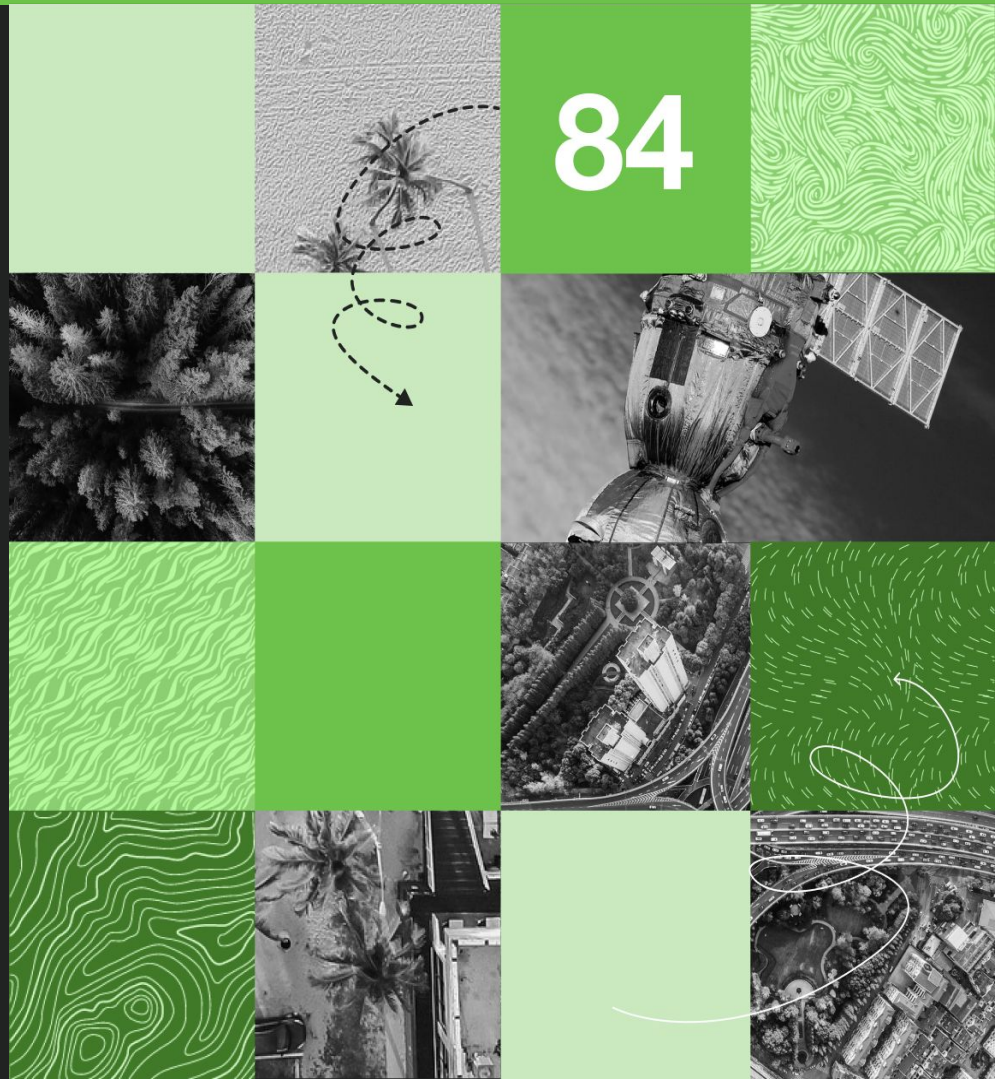
What AI Coding means for the future of Open-Source Geospatial Software

FOSS4G | Auckland, NZ | Nov 2025

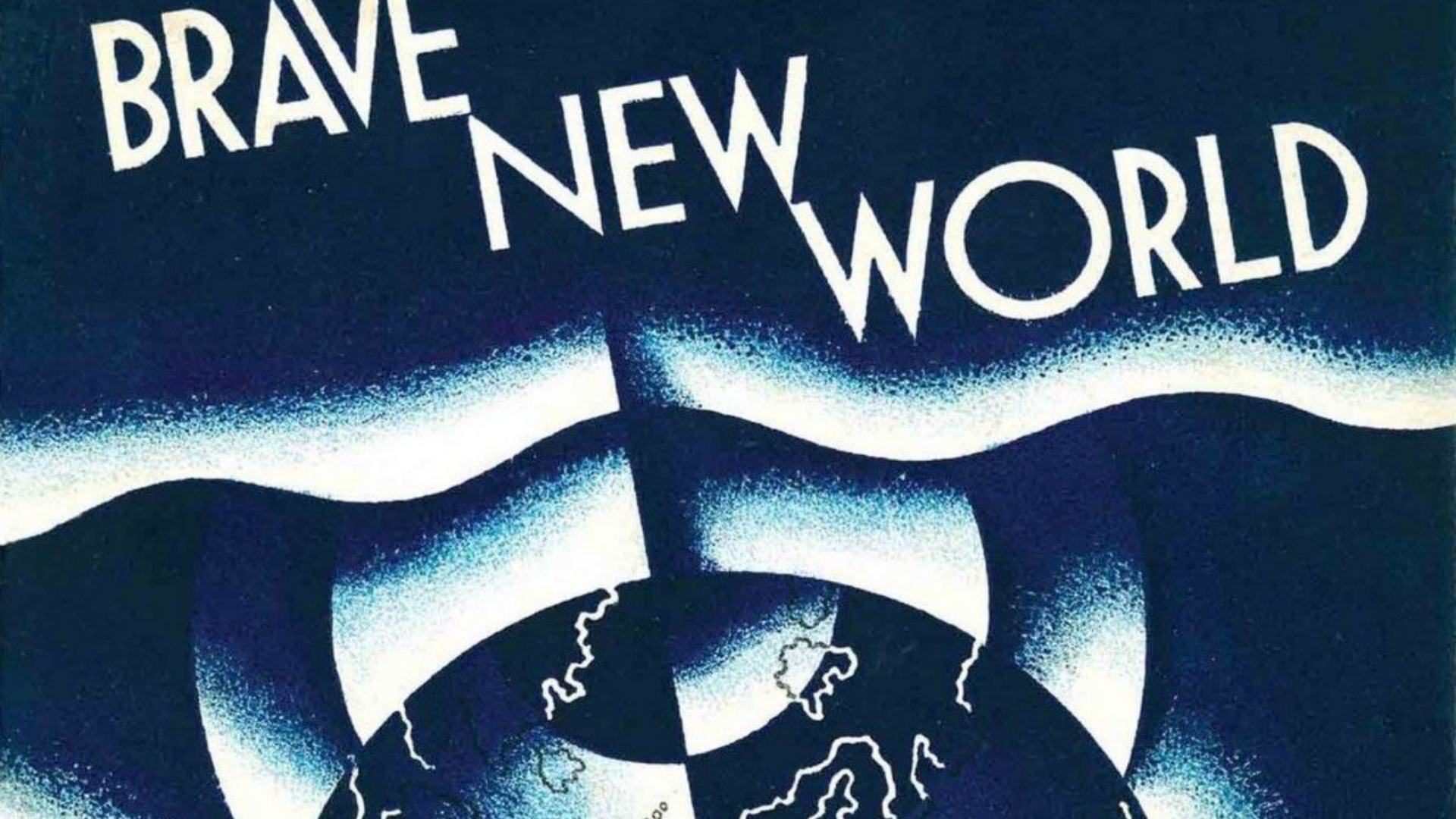


Matthew Hanson
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Element 84

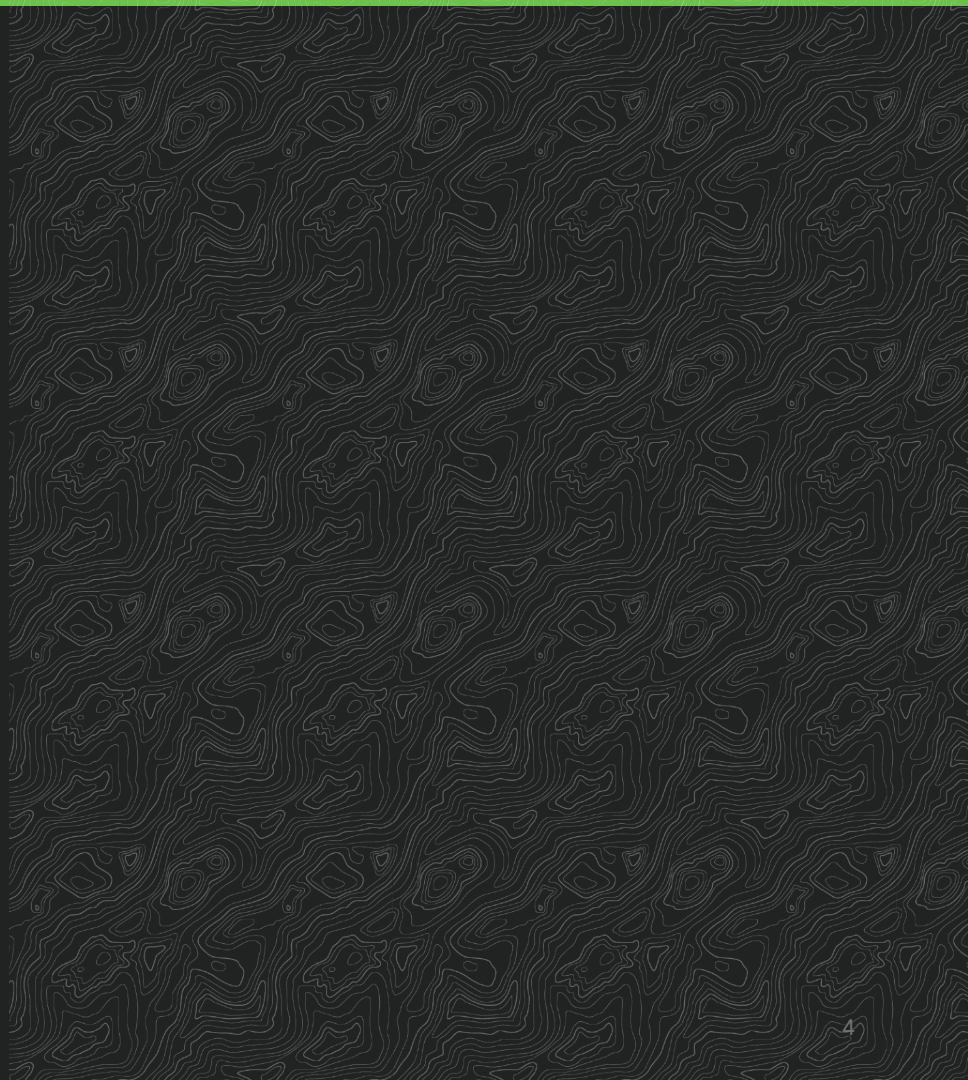


BRAVE NEW WORLD





Why this matters




FREE
OR THE UNEMPLOYED
FREE SOUP
&
PARKING 25
ALBERT
HORAN
BAILIFF

The Fork in the Code

- AI is risking greater fragmentation in Open-Source software
- Subject matter expertise, such as geospatial, matters more
- AI “completeness” is shallow and bypasses real libraries.
- We must stabilize the existing geospatial foundations
- We must move up the stack and define the new foundations AI will build upon



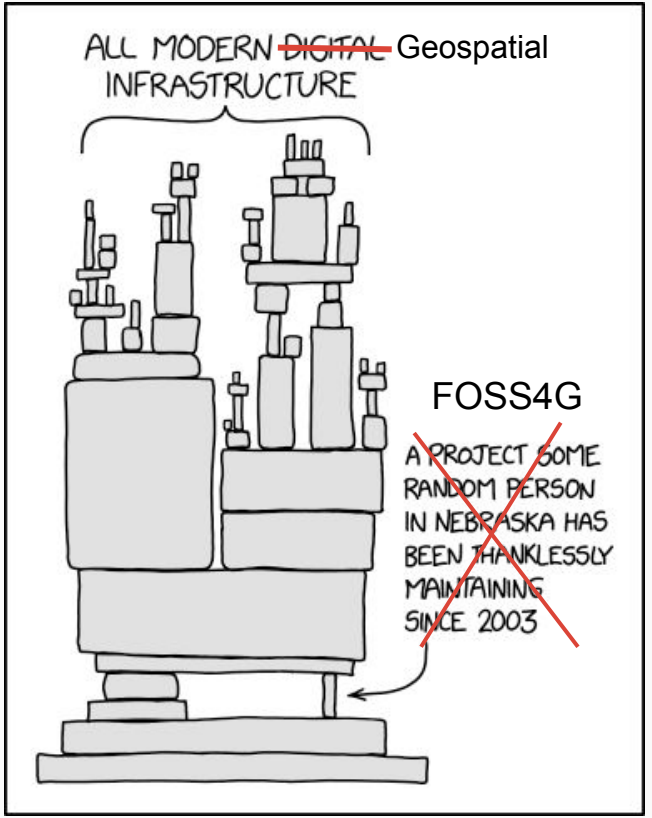
How did we get here?

The Timeline

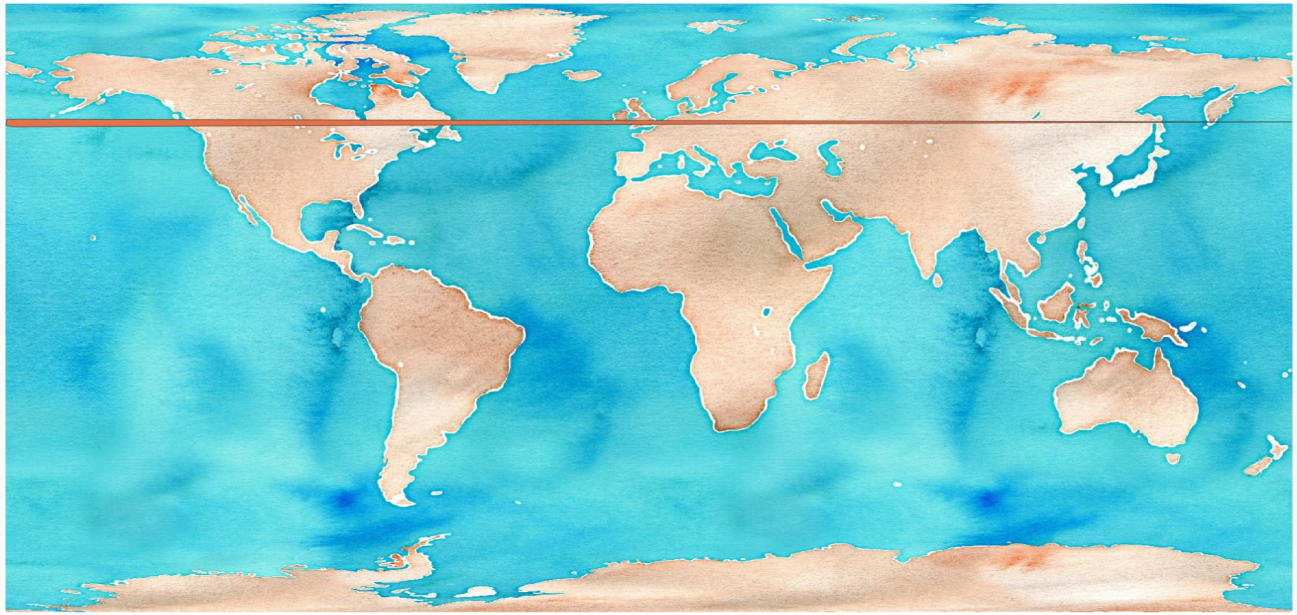
- **2021**: GitHub Copilot launches (technical preview).
- **2022**: ChatGPT publicly released.
- **2023**: GPT-4 arrives; coding assistants gain broader attention.
- **2023-24**: LLM’s scaffold new repos
- **2024:** Specialization of models: for whole repo tasks, brainstorming, or quick coding
- **2024–25**: Long-context models (e.g., GPT-4.1) enable whole-repo reasoning
- **2025**: SpecKit and the spec-driven shift emerge.

The Geospatial Isn’t Special Paradox

- “Geo not special” driven integration w/ general data infrastructure
- Successes like GeoParquet rely on shared domain expertise
- A small maintainer base supports a huge global user base.
- AI has no domain grounding.
    - It treats geospatial as generic.
    - It produces brittle, vibe-coded solutions.



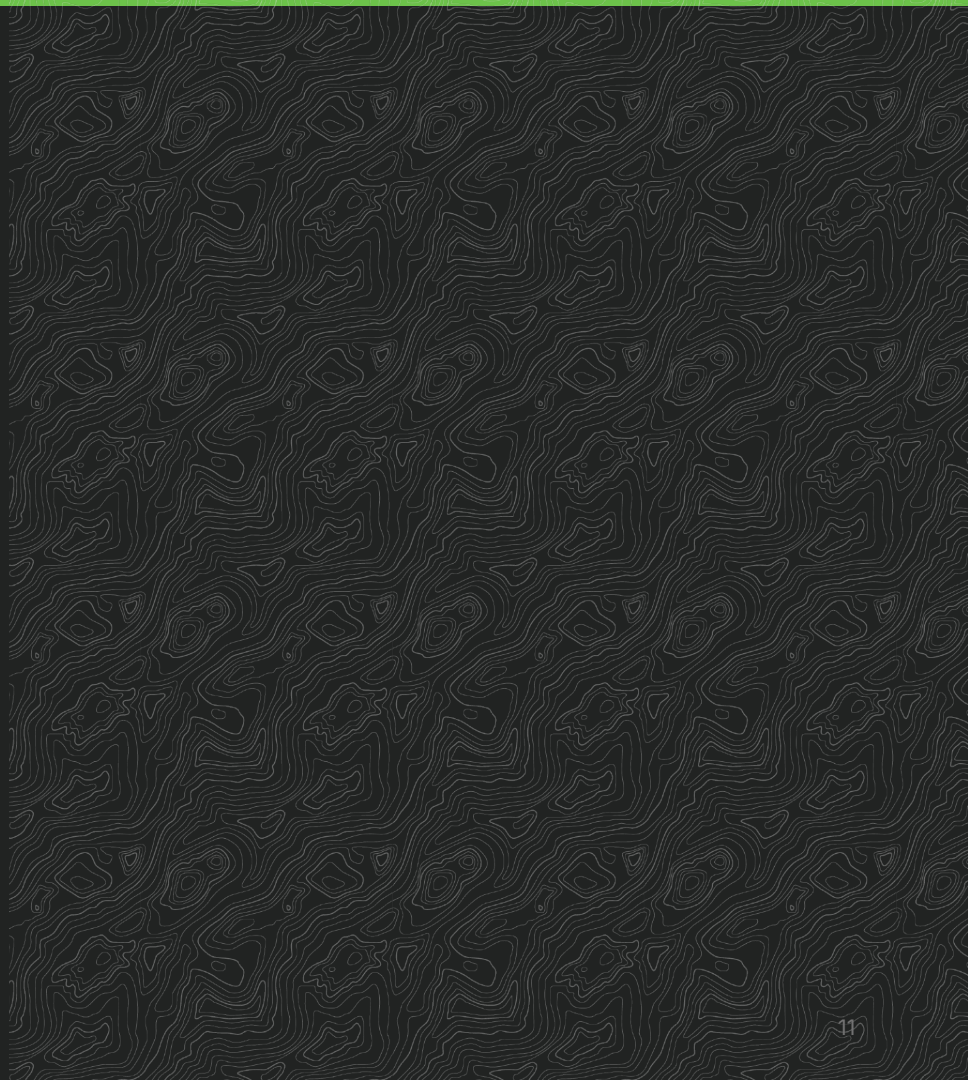
antimeridian



https://github.com/gadomski/antimeridian

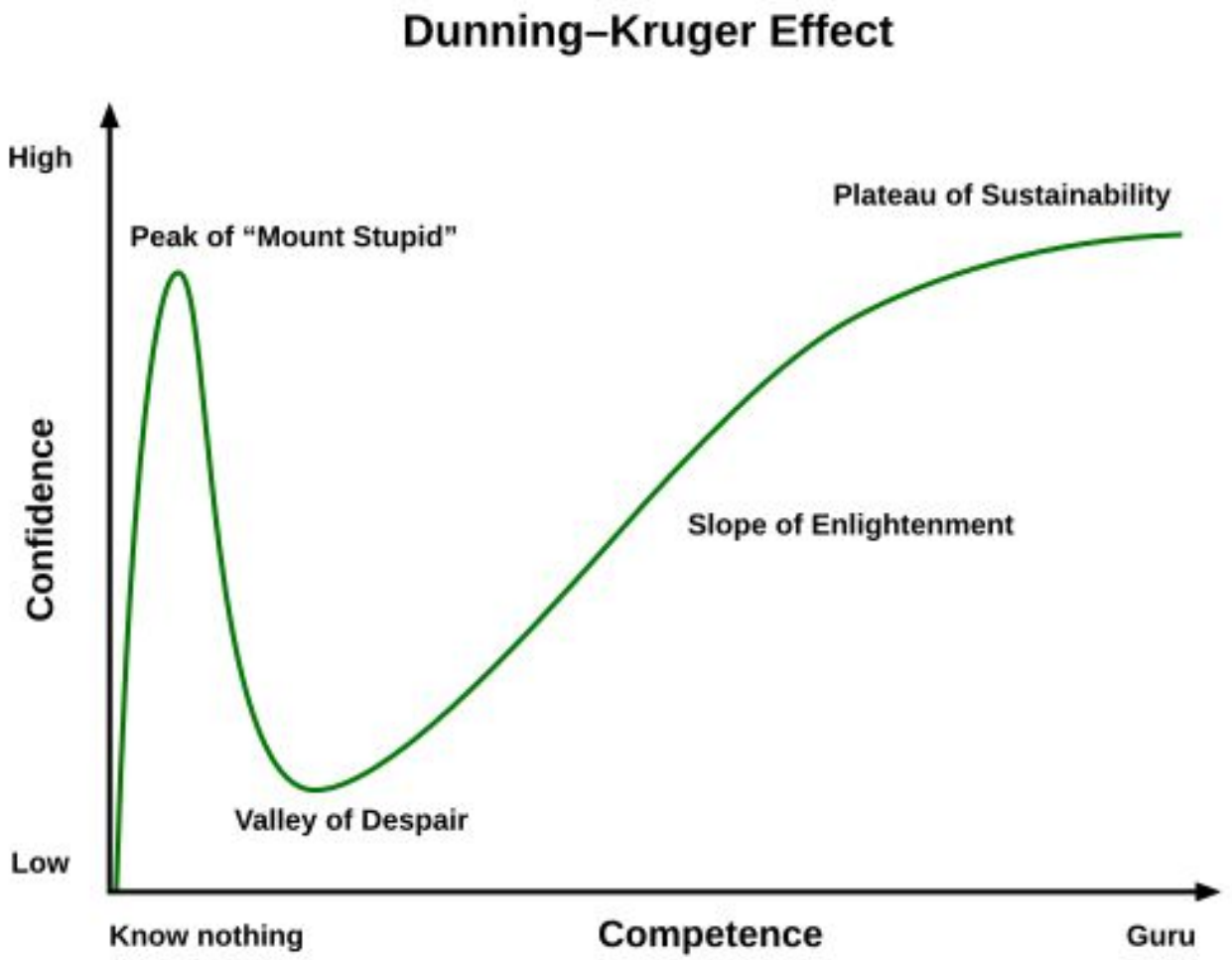
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            ]
        ]
    ]
}
```

Fragmentation



The Illusion of Completeness




Dunning–Kruger Effect
High
Confidence
Low
Peak of "Mount Stupid"
Plateau of Sustainability
Slope of Enlightenment
Valley of Despair
Know nothing
Competence
Guru

The broken discovery process



OGC API - Features Implementations

Servers

The columns for each part list the conformance classes of the standard that are implemented by the server implementation. The conformance classes available in a specific API that is provided using the implementation will be listed in the Conformance Declaration resource of the API.

Table 1. Server implementations

Product	Part 1	Part 2	Part 3	OGC Product Database	Contact email
ldproxy	`core`, `oas30`, `geojson`, `html`, `gmlsf2`	`crs`	`filter`, `features-filter`, `simple-cql`, `cql-json`, `cql-text`	Link	portele [at] interactive-instruments.de
CubeWerx Suite	`core`, `oas30`, `geojson`, `html`, to be updated	`crs`	`filter`, `features-filter`, `simple-cql`, `cql-json`, `cql-text`, to be updated	Link	pvretano [at] cubewerx.com
GeoServer	please consult the product documentation for details			Link	
pygeoapi	please consult the product documentation for details			Link	
sofp Server	please consult the product documentation for details			Link	
hakunapi	`core`, `oas30`, `geojson`	`crs`	-	-	
QGIS Server	please consult the product documentation for details			Link	

What CTOs Really Think About Vibe Coding

We asked 18 CTOs about vibe coding. 16 reported production disasters. Here's why AI-generated code is creating more problems than it solves for real engineering teams.



Written by
Kaustubh Saini



Edited by
Jaya Muvania



Reviewed by
Kaivan Dave

Updated on
Aug 15, 2025

Read time
7 Min Read

Amateurs Using AI to "Vibe Code" Are Now Begging Real Programmers to Fix Their Botched Software

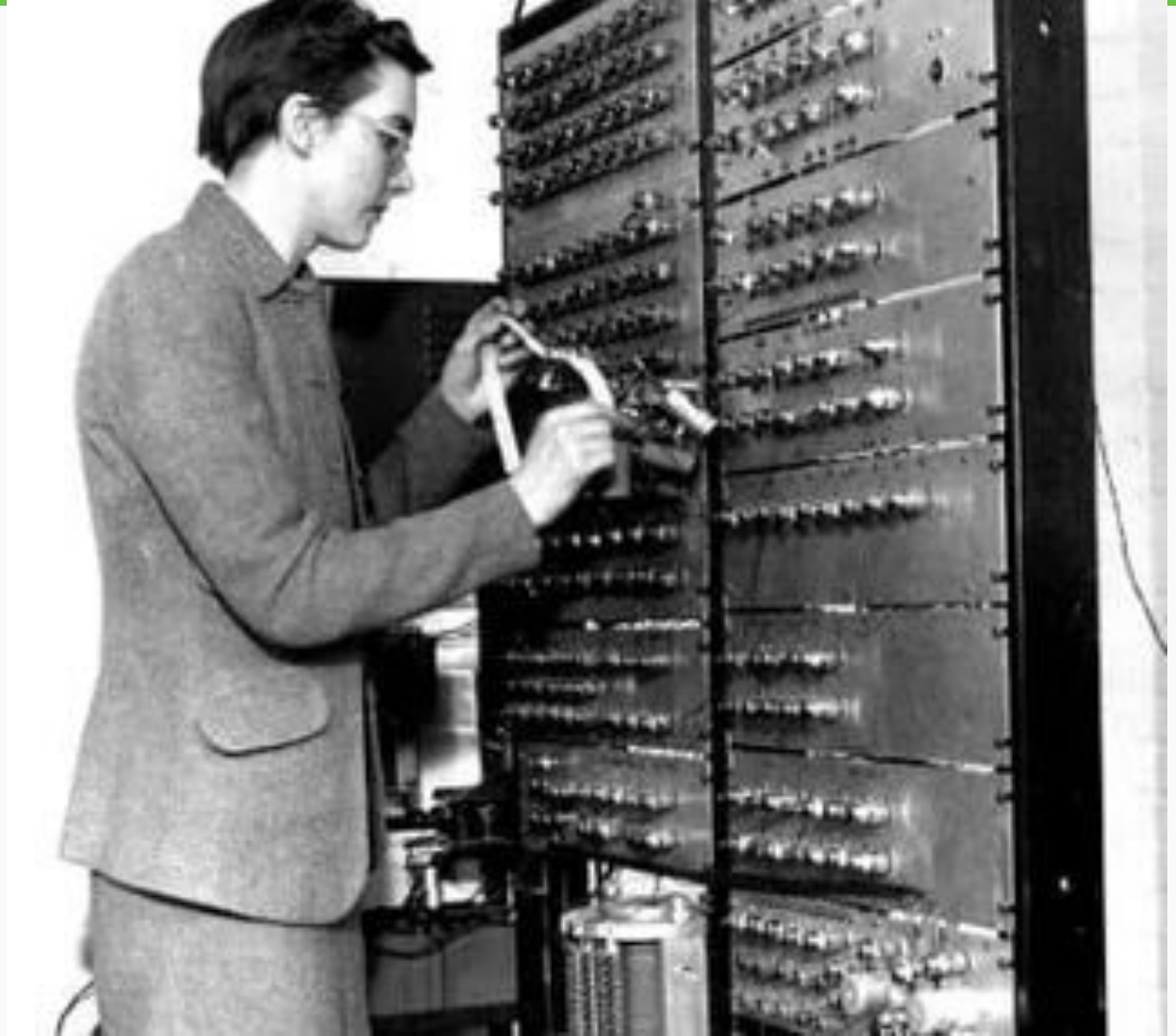
Vibe first, ask for professional help later.



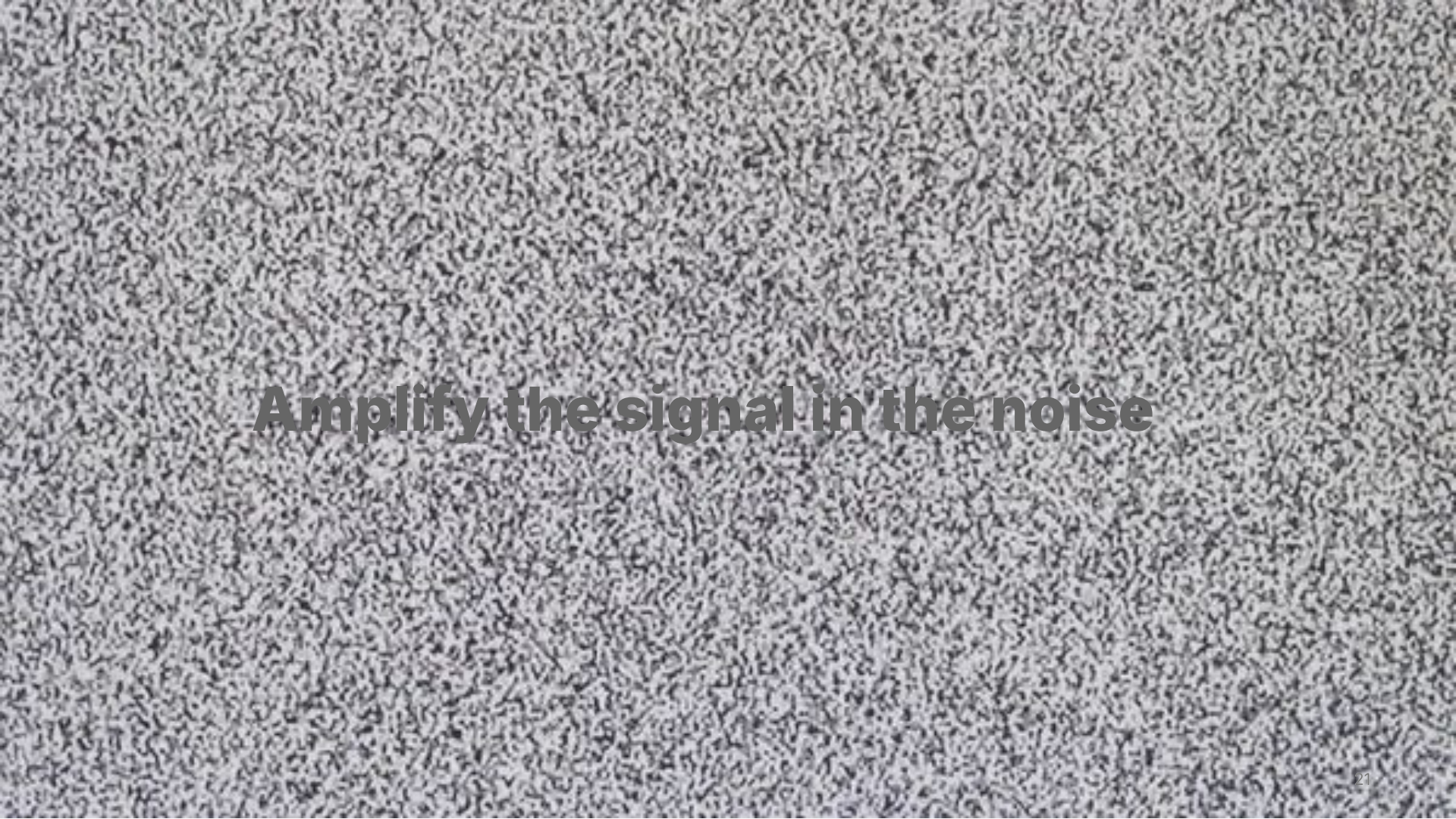
By **Frank Landymore** / Published **Sep 16, 2025 2:49 PM EDT**

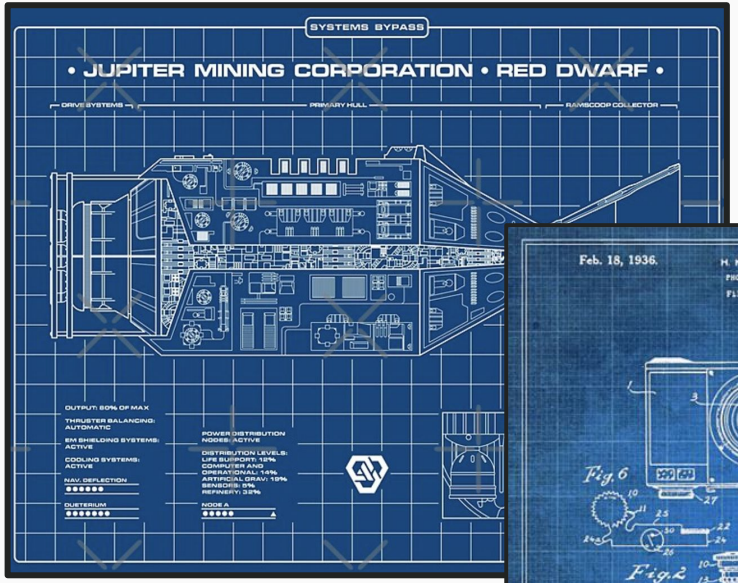


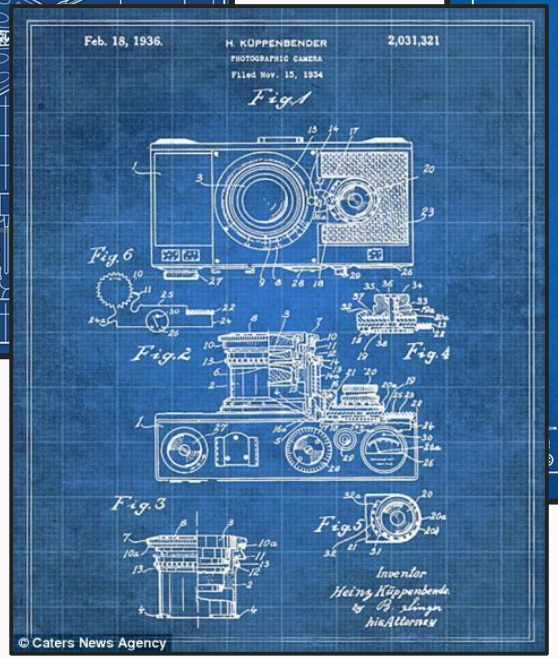
We’ve been here before

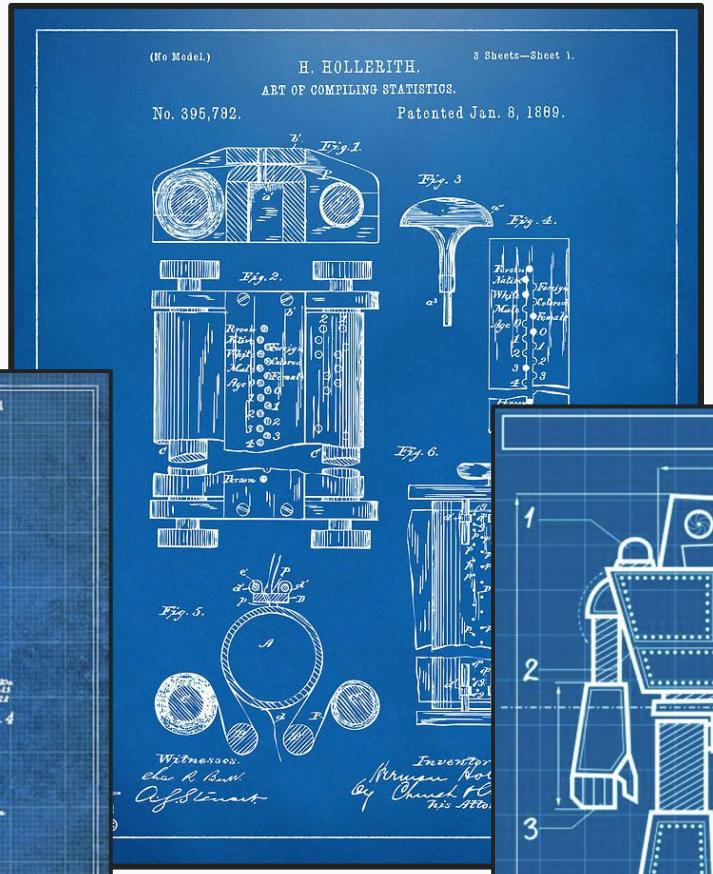


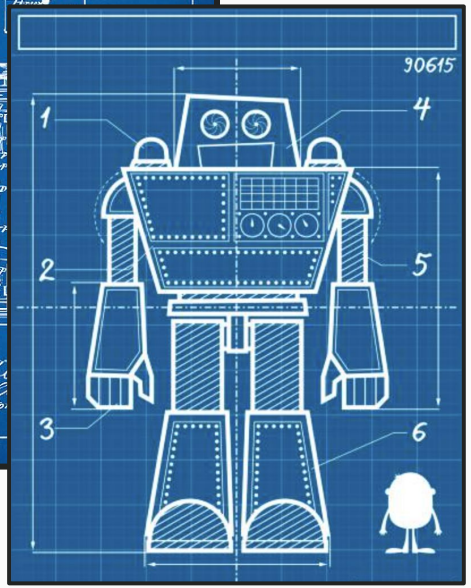
Now & Next


Amplify the signal in the noise





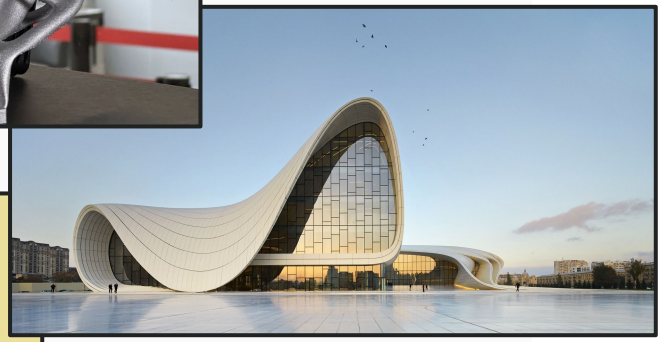


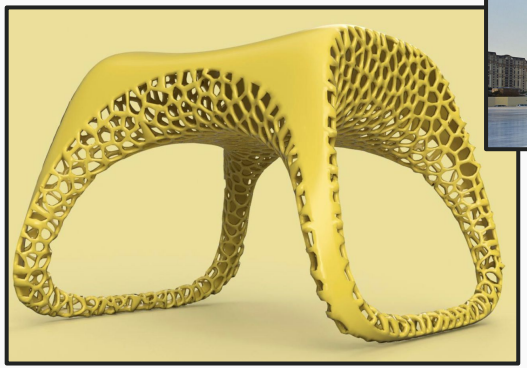


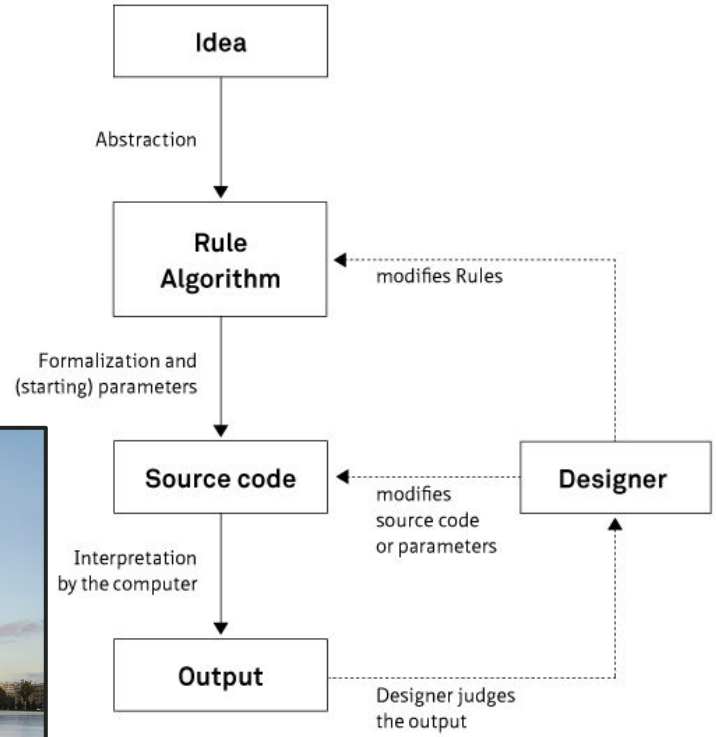
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Generative Design









Copyright Hartmut Bohnacker, Julia Laub, Benedikt Groß, Claudius Lazzeroni (2009)
Book „Generative Gestaltung", www.generative-gestaltung.de

Takeaways

Takeaways

- AI Creates an "Illusion of Completeness"
- Fragmentation is the Default
- Subject Matter Experts Become More Valuable, Not Less
- The new "Compiler" Is Here, So We Must Act
    - **Now** - Amplify the signal by improving documentation and tutorials
    - **Next** - Build the foundry, the new ecosystem we will use as designers

Conclusion


SYSTEM FAILURE










WALL·E

Thank you!

Let’s discuss



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Element 84









